

Task 1:

List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Answer:

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Uses decimal format for addresses.	Uses hexadecimal format for addresses.
Example: 192.168.1.1	Example: 2001:db8::1
Provides about 4.3 billion addresses.	Provides a very large number of addresses.
Uses dotted-decimal notation.	Uses colon-separated hexadecimal notation.
NAT is commonly used due to limited addresses.	NAT is generally not required due to the large address space.
Supports broadcast communication.	Does not support broadcast; it uses multicast and anycast.
Uses ARP to find the MAC address of a device.	Uses Neighbor Discovery Protocol (NDP) instead of ARP.
Uses DHCP or manual configuration.	Supports SLAAC and DHCPv6 for address configuration.
Header size is 20–60 bytes.	Header size is 40 bytes and has a fixed format.

Task 2:

Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Answer:

For the home WiFi network 192.168.10.0/24, we can assign unique static IPv4 addresses to the four devices as follows:

1. Laptop

- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

2. Phone

- IP Address: 192.168.10.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

3. Smart TV

- IP Address: 192.168.10.12
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

4. Printer

- IP Address: 192.168.10.13
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Here, 192.168.10.0 is the network address, 192.168.10.1 is used as the default gateway, and each device gets a different IP address within the same network.

Task 3:

Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Answer:

For the network 192.168.1.0/24, if we use a /27 CIDR subnet mask:

- Subnet mask: 255.255.255.224
- Borrowed bits: $27 - 24 = 3$ bits
- Number of subnets: $2^3 = 8$ valid subnets
- Host bits: $32 - 27 = 5$ bits
- Addresses per subnet: $2^5 = 32$
- Usable hosts per subnet: $32 - 2 = 30$ hosts
- Subnet increment: $256 - 224 = 32$

First 3 valid subnets

1. Subnet: 192.168.1.0/27

- Network address: 192.168.1.0
- Broadcast address: 192.168.1.31
- Usable host IPs: 192.168.1.1 – 192.168.1.30

2. Subnet: 192.168.1.32/27

- Network address: 192.168.1.32
- Broadcast address: 192.168.1.63
- Usable host IPs: 192.168.1.33 – 192.168.1.62

3. Subnet: 192.168.1.64/27

- Network address: 192.168.1.64
- Broadcast address: 192.168.1.95
- Usable host IPs: 192.168.1.65 – 192.168.1.94

Final answer: A /27 mask creates 8 subnets, with 30 usable hosts per subnet.

Task 4:

You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Answer:

Given IP address: 10.0.5.17

Subnet mask: 255.255.255.0

Step 1: Convert subnet mask to CIDR

255.255.255.0 = /24

So, the network is 10.0.5.0/24.

Step 2: Find Network Address

Using bitwise AND:

IP: 10.0.5.17

Mask: 255.255.255.0

Network: 10.0.5.0

Step 3: Find Broadcast Address

For a /24 network, the last 8 bits are available for hosts.

Therefore, the broadcast address is:

10.0.5.255

Step 4: Valid Host Range

The network address 10.0.5.0 and broadcast address 10.0.5.255 cannot be assigned to normal hosts.

So, the valid host IP range is:

10.0.5.1 – 10.0.5.254

Final Answer

- CIDR: /24
- Network Address: 10.0.5.0
- Broadcast Address: 10.0.5.255
- Valid Host Addresses: 10.0.5.1 – 10.0.5.254
- Usable Hosts: 254

Task 5:

Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Answer:

Subnet Calculator

Network Class: A ☐ B ☐ C ☒

IP Address: 192.0.5.17

Subnet Mask: 255.255.255.0

Subnet Bits: 0

Maximum Subnets: 1

Host Address Range: 10.0.5.1 - 10.0.5.254

Subnet ID: 10.0.5.0

Subnet Bitmap: 110nnnnn.nnnnnnnn.nnnnnnnn.hhhhhhhh

First Octet Range: 192 - 223

Hex IP Address: 0A.00.05.11

Wildcard Mask: 0.0.0.255

Mask Bits: 24

Hosts per Subnet: 254

Broadcast Address: 10.0.5.255

IP Subnet Calculator

The IP Subnet Mask Calculator enables subnet network calculations using network class, IP address, subnet mask, subnet bits, mask bits, maximum required IP subnets and maximum required hosts per subnet.

Results of the subnet calculation provide the hexadecimal IP address, the wildcard mask, for use with ACL (Access Control Lists), subnet ID, broadcast address, the subnet address range for the resulting subnet network and a subnet bitmap.

For classless supernetting, please use the CIDR Calculator. For simple ACL (Access Control List) wildcard mask calculations, please use the ACL Wildcard Mask Calculator.

My manual calculation matched the subnet calculator output: Network Address = 10.0.5.0, Broadcast Address = 10.0.5.255, and Valid Host Range = 10.0.5.1–10.0.5.254.